

Hasan Zakeri

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SUMMARY

Machine learning engineer with 8+ years of experience building LLM-based agentic systems, evaluation frameworks, and safety-critical AI pipelines. Recent work spans agentic document processing at scale under GDPR/HIPAA and attorney-client privilege constraints, MCP server architecture, and LLM observability. Ph.D. research in deep reinforcement learning and Bayesian uncertainty quantification for autonomous systems, published in IEEE and ASME. Production deployment experience across CI/CD, containerized infrastructure, and scalable data tooling.

SKILLS AND CERTIFICATES

Machine Learning: LLM finetuning, RAG, agentic workflows, MCP, reinforcement learning, deep learning, transformers, graph neural networks, Bayesian uncertainty quantification.

Programming: Python (PyTorch, Scikit-Learn, NumPy, Pandas), C/C++, SQL, Django.

Infrastructure & MLOps: Docker, Prefect, CI/CD, AWS, vector databases, Spark, ETL pipelines.

EDA & Simulation Tools: Matlab/Simulink, CARLA, pyChrono, PyBullet, Airsim, CarSim.

Certificates: Designing Microsoft Azure Infrastructure Solutions (AZ-305), Implementing Analytics Solutions Using Microsoft Fabric (DP-600), Generative AI with Large Language Models, Amateur Radio Extra Class

EDUCATION

Illinois Institute of Technology

Chicago, IL

Ph.D. in Mechanical and Aerospace Engineering

Expected 2026

Available for full-time roles immediately

University of Notre Dame

Notre Dame, IN

M.S. in Electrical Engineering (Advisor: Panos Antsaklis)

EXPERIENCE

Johnson Law Group, AI Engineering Intern

Remote, Feb–Aug 2026

- Designed end-to-end agentic workflows for LLM-driven legal document processing, absorbing a burst of 400,000 documents, largely scanned and faxed records requiring identification and splitting before downstream processing, with systematic evaluation of outputs against defined accuracy and reliability benchmarks.
- Eliminated manual document extraction from a workflow previously handled by human readers; defined confidence metrics for pipeline outputs and tuned against them, reducing the share of documents requiring human review by 90%.
- Architected an MCP server with web search and natural language data lake querying tools, engineering strict safeguards against PHI/PII leakage across a deployed chat interface and downstream agentic pipelines.
- Defined and enforced data handling practices in code, routing each class of data only to vendors and models permitted under its data-processing agreement, ensuring GDPR, HIPAA, and attorney-client privilege compliance.
- Built observability tooling, including an LLM cost-tracking ledger, to monitor pipeline reliability, cost, and model behavior at scale.
- Overhauled Prefect orchestration and migrated test/CI to containerized Prefect-Docker deployments, boosting reproducibility.

Illinois Institute of Technology, Graduate Research Assistant

Chicago, IL, Ongoing

- Developed a deep learning framework substantially improving uncertainty quantification for autonomous vehicle state prediction, using Bayesian neural networks and Gaussian process regression.
- Built a multi-agent reinforcement learning framework improving traffic flow by 40% and reducing congestion time by 30%.

University of Notre Dame, Graduate Research Assistant

Notre Dame, IN, 2016–2019

- Designed data-driven resilient controller achieving 95% attack detection accuracy using unsupervised machine learning.
- Developed polynomial approximation framework reducing computational cost with closed-loop performance guarantees.

EmNet LLC., Machine Learning Research Intern

South Bend, IN, 2016

- Designed a deep learning architecture for urban sewer flow prediction, validated across 4 major US cities.
- Implemented a neural network controller in C for EPA-SWMM, deployed in production across multiple municipalities.

EARLIER EXPERIENCE

Tachara Online Bookstore, Technical Project Lead

Tehran, Iran, 2010–2013

- Designed an e-commerce platform using .NET MVC 4, C# and SQL Server serving over 1M active users; led integration of payment gateways and inventory systems.

Azar Negar Shargh Co., Principal Engineer

Shiraz, Iran, 2004–2009

- Architected full-stack content management systems and led cross-functional engineering teams for corporate communications clients.

MACHINE LEARNING AND CONTROL PROJECTS

Traffic Congestion using Deep RL | PyTorch, PPO, Multi-Head Attention, Actor-Critic Networks, GNN

Ongoing

- Attention-based PPO agent to optimize highway traffic flow through speed commands, achieving 25–40% congestion reduction
 - Designed using multi-head attention state to handle 10–100+ vehicles with lane-change and graph-based highway topology
- Distributed Filter Design Agent** | *Vision Transformers, YOLO, ResNet, SFT, ADS, HFSS* Winter 2024
- Trained multi-modal Vision Transformer with YOLO and ResNet on filter topologies with 92% accuracy after post-training.
 - Developed end-to-end AI agent for synthesizing novel filters from desired characteristics, collapsing a manual design-iteration loop into a single automated pass.
- RF Signal Classification Assistant** | *PyTorch, Transformers, Signal Processing* Spring 2024
- Developed multimodal LLM combined with spectral analysis to classify and characterize unknown RF signals, achieving 95% classification accuracy.
 - Built conversational interface to query signal characteristics using natural language, reducing analysis time by 50%.
- Autonomous Vehicles Control with RL** | *CARLA simulator, Stable Baselines, PyTorch, OpenCV* Spring 2023
- Trained a reinforcement learning agent using CARLA simulator images to autonomously control vehicle navigation.
 - Developed a vision-based controller that learned driving in a realistic urban environment without explicit programming.
- Vehicle Dynamics Model: Uncertainty Quantification** | *pyChrono, scikit, PyTorch, torchbnn* Winter 2022
- Validated first-principles vehicle dynamics models against Chrono::Vehicle simulations across steering and acceleration maneuvers.
 - Characterized modeling uncertainties using Gaussian Process Regression and Bayesian neural networks, quantifying prediction confidence bounds against simulation ground truth.
- RF Filter Synthesis Optimization** | *Python, scikit-rf, NumPy, SciPy, Matplotlib, MATLAB* Summer 2022
- Built numerical optimization pipeline for filter synthesis and component selection, parsing manufacturer touchstone and pspice files into a searchable design space.
 - Developed automated response-matching between target and simulated filter characteristics.
- LCR Meter Design Using Auto-Balancing Bridge Topology** | *STM32, Altium Designer, Analog Design, PLL* Spring 2022
- Designed and implemented an impedance analyzer using auto-balancing bridge topology for RF component measurements.
 - Achieved measurement accuracy of close to $\pm 0.5\%$ over frequency range of 100 Hz to 10 MHz.

SELECTED PUBLICATIONS

- **Hasan Zakeri** and Baisravan HomChaudhuri, “Graph Neural Network and Attention-Based Deep Reinforcement Learning for Heterogeneous Traffic Congestion Control with Variable Fleet Dynamics,” In preparation for The International Conference on Machine Learning (ICML 2027).
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Learning-Based Predictive Control Method for Vehicle Lateral Control with a Multi-Step Gaussian Regression Prediction,” submitted, *Journal of Dynamic Systems, Measurement, and Control*, 2025.
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Iterative Stochastic Model Predictive Control with Multi-Step Uncertainty Learning Model: Vehicle Lane Change Case Study,” *Modeling, Estimation and Control Conference, MECC* 2024.
- **Hasan Zakeri** and Baisravan HomChaudhuri, “Multi-Step Gaussian Regression Prediction in Dynamic Systems: A Case Study in Vehicle Lateral Control,” in *ASME Letters in Dynamic Systems and Control* vol. 3, no. 4, Oct. 2023.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity Measures in Cyberphysical Systems Design: An Overview of Recent Results and Applications,” in *IEEE Control Systems Magazine*, vol. 42, no. 2, pp. 118-130, April 2022.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity and Passivity Indices for Switched and Cyber-Physical Systems,” (invited) in *Encyclopedia of Systems and Control*, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity, Dissipativity and Passivity Indices,” (invited) in *Encyclopedia of Systems and Control*, J. Baillieul and T. Samad, Eds. London: Springer London, 2020.
- **Hasan Zakeri** and Panos J. Antsaklis, “Passivity and Passivity Indices of Nonlinear Systems under Operational Limitations using Approximations,” *International Journal of Control*, July 2019.
- **Hasan Zakeri** and Panos J. Antsaklis, “A Data-driven Adaptive Controller Reconfiguration for Fault Mitigation: A Passivity Approach,” 27th Mediterranean Conference on Control and Automation (MED), Akko, Israel, 2019, pp. 25–30.
- More at <https://scholar.google.com/citations?user=a-Z05tMAAAAJ>

LEADERSHIP AND SERVICES

- Search committee, Graduate Career Director, Notre Dame — reviewed and interviewed 6 candidates. January 2018
- Professional Development Chair, Graduate Student Union, University of Notre Dame 2017–2018
- University Committee on Internationalization, appointed by the President, University of Notre Dame. 2017–2018
- Jury and Test Designer for Iran’s National Skills Olympiad on IT Software Solutions for Business 2013